caculation engine

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Chapter 1

Calculation Engine Documentation

1.1 Introduction

This project demonstrates a simple C++ calculation engine with logging functionality.

1.2 Features

- Displays informational log messages.
- Calculates the power of a number.
- Uses Doxygen comments for documentation.

Chapter 2

Todo List

Class CalculationEngine

Add more mathematical operations in the future.

Chapter 3

Class Index

3.1 Class List

Here are the classes, structs, unions and interfaces with brief descriptions:

CalculationEngine
Performs mathematical calculations 9
Logger
Provides logging functionality 10

Chapter 4

File Index

4.1 File List

Here is a list of all files with brief descriptions:

calculationengine.h
: Defines the CalculationEngine class 13

logger.h
: Defines the Logger class and logging levels 14

main.cpp
: Entry point of the calculation application 15

Chapter 5

Class Documentation

5.1 CalculationEngine Class Reference

Performs mathematical calculations.

`#include <calculationengine.h>`

Public Member Functions

- CalculationEngine (const Logger &log)
  Constructs a CalculationEngine object.
- double power (double base, int exponent) const
  Calculates the power of a number.

5.1.1 Detailed Description

Performs mathematical calculations.

This class provides mathematical operations and uses a Logger object to display calculation-related messages.

Todo Add more mathematical operations in the future.

5.1.2 Constructor & Destructor Documentation

5.1.2.1 CalculationEngine()

```
CalculationEngine::CalculationEngine (
            const Logger & log)  [inline]
```

Constructs a CalculationEngine object.

Parameters

log	A Logger object used for logging.

5.1.3 Member Function Documentation

5.1.3.1 power()

```
double CalculationEngine::power (
            double base,
            int exponent) const  [inline]
```

Calculates the power of a number.

Parameters

base	The number to be raised to a power.
exponent	The power to which the base is raised.

Returns

The calculated result.

The documentation for this class was generated from the following file:

- calculationengine.h

5.2 Logger Class Reference

Provides logging functionality.

```
#include <logger.h>
```

Public Member Functions

- Logger (const std::string &loggerName)
  Constructs a Logger object.
- void log (LogLevel level, const std::string &message) const
  Displays a log message.

5.2.1 Detailed Description

Provides logging functionality.

The Logger class displays messages with their corresponding logging levels and component name.

5.2.2 Constructor & Destructor Documentation

5.2.2.1 Logger()

```
Logger::Logger (
            const std::string & loggerName)  [inline]
```

Constructs a Logger object.

Parameters

loggerName	Name of the component using the logger.

5.2.3 Member Function Documentation

5.2.3.1 log()

```
void Logger::log (
            LogLevel level,
            const std::string & message) const  [inline]
```

Displays a log message.

Parameters

level	The severity level of the message.
message	The message to be displayed.

The documentation for this class was generated from the following file:

- logger.h

Chapter 6

File Documentation

6.1 calculationengine.h File Reference

Defines the CalculationEngine class.

```
#include "Logger.h"
#include <cmath>
```

Classes

- class CalculationEngine

  Performs mathematical calculations.

6.1.1 Detailed Description

Defines the CalculationEngine class.

6.2 calculationengine.h

Go to the documentation of this file.

```
00001
00005
00006 #ifndef CALCULATION_ENGINE_H
00007 #define CALCULATION_ENGINE_H
00008
00009 #include "Logger.h"
00010 #include <cmath>
00011
00022 class CalculationEngine
00023 {
00024 private:
00025     Logger logger;
00026
00027 public:
00028
00033     CalculationEngine(const Logger& log)
00034         : logger(log)
00035     {
00036     }
00037
00044     double power(double base, int exponent) const
00045     {
00046         logger.log(LogLevel::INFO, "Calculating power");
00047
00048         return std::pow(base, exponent);
00049     }
00050 };
00051
00052 #endif
```

6.3 logger.h File Reference

Defines the Logger class and logging levels.

```
#include <iostream>
#include <string>
```

Classes

- class Logger

  Provides logging functionality.

Enumerations

- enum class LogLevel { INFO , WARNING , ERROR }

  Defines the different levels of logging.

6.3.1 Detailed Description

Defines the Logger class and logging levels.

This file provides functionality to display informational, warning, and error messages.

6.3.2 Enumeration Type Documentation

6.3.2.1 LogLevel

```
enum class LogLevel  [strong]
```

Defines the different levels of logging.

Enumerator

INFO	Informational messages.
WARNING	Warning messages.
ERROR	Error messages.

6.4 logger.h

Go to the documentation of this file.

```
00001
00009
00010 #ifndef LOGGER_H
00011 #define LOGGER_H
00012
00013 #include <iostream>
00014 #include <string>
00015
00020 enum class LogLevel
00021 {
00022     INFO,
00023     WARNING,
00024     ERROR
00025 };
00026
00034 class Logger
00035 {
00036 private:
00037     std::string name;
00038
00039 public:
00040
00045     Logger(const std::string& loggerName)
00046         : name(loggerName)
00047     {
00048     }
00049
00055     void log(LogLevel level, const std::string& message) const
00056     {
00057         std::string levelName;
00058
00059         switch (level)
00060         {
00061             case LogLevel::INFO:
00062                 levelName = "INFO";
00063                 break;
00064
00065             case LogLevel::WARNING:
00066                 levelName = "WARNING";
00067                 break;
00068
00069             case LogLevel::ERROR:
00070                 levelName = "ERROR";
00071                 break;
00072         }
00073
00074         std::cout « "[" « levelName « "] "
00075                   « name « ": "
00076                   « message « std::endl;
00077     }
00078 };
00079
00080 #endif
```

6.5 main.cpp File Reference

Entry point of the calculation application.

```
#include "Logger.h"
#include "CalculationEngine.h"
#include <iostream>
```

Functions

- int main ()

6.5.1 Detailed Description

Entry point of the calculation application.

This program demonstrates the use of Logger and CalculationEngine classes.

6.5.2 Function Documentation

6.5.2.1 main()

```
int main ()
```

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